Newton

A physics visualizer for the classroom

High-Up! Hackathon • 19–20 November

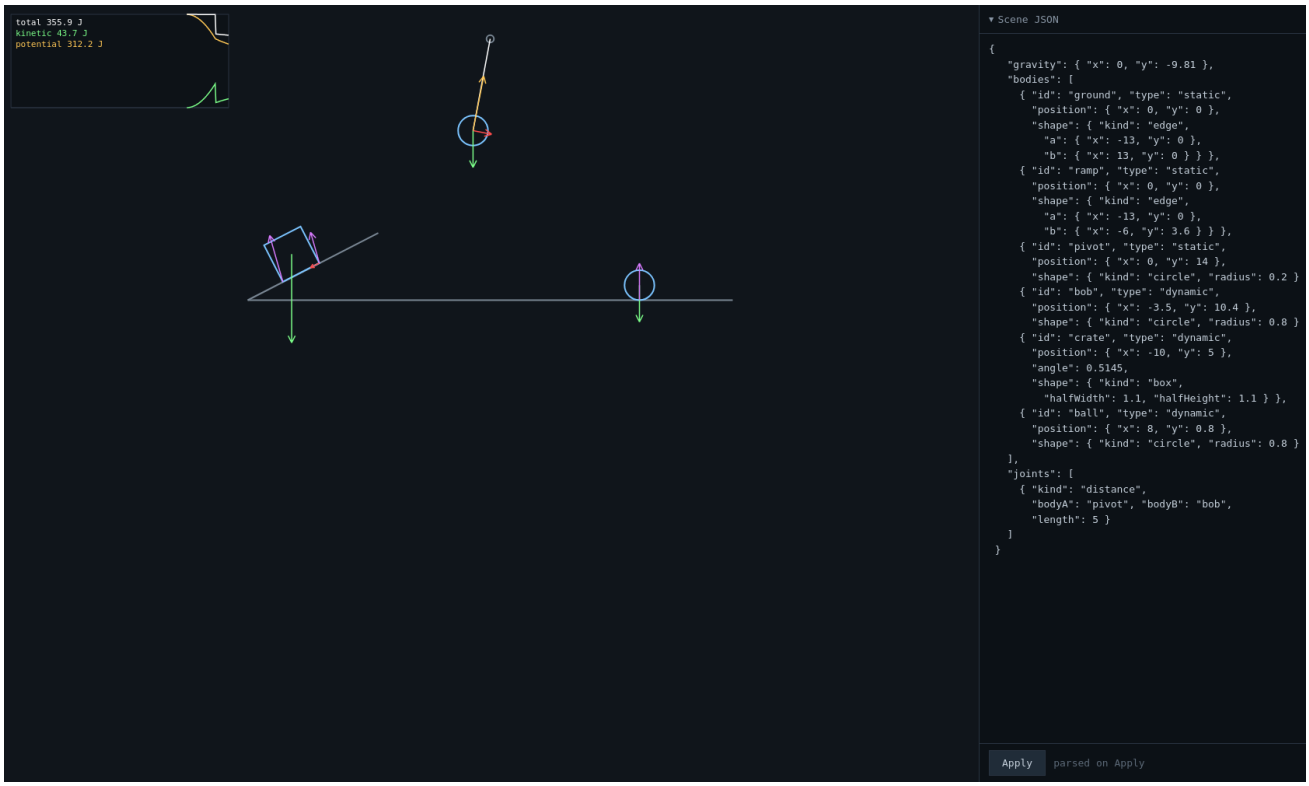
Physics teachers spend hours drawing the same free-body diagrams, and students still cannot see the forces changing while the system moves. Newton turns a plain-language description of a physical system into a live, time-evolving visualization: bodies, joints, the forces acting on each body, and the energy budget as it unfolds.

The problem

- **Diagrams freeze time.** A static free-body diagram shows one instant. Students memorize arrows instead of seeing *when* and *why* they change.
- **Simulation tools are expert tools.** Existing physics engines want code, scene files and a build step; PhET-style applets are polished but closed — you cannot ask for *your* problem, the exact ramp angle from chapter 7.
- **Teachers have no fast path from a problem statement to a visual.** Preparing a demo is a separate project from preparing the lesson.

Newton targets the gap: **expressive** enough for a specific textbook exercise, **fast** enough to use live in class, and **reliable** enough that a wrong answer is never silently displayed.

How it works



Current build, running. A pendulum released at an angle, a crate resting on a ramp, and a ball on the ground. Green arrows are gravity, ochre the joint reaction, magenta the normal force from the ramp and ground, red the velocity; the panel top-left plots kinetic, potential and total energy over the last 5 seconds. The right-hand pane is the scene document being edited.

A scene is JSON describing bodies (`circle`, `box`, `edge`), their mass, shape and pose, and joints between them — all SI units, $+y$ up, anchors local to their body so a joint stays attached when the body moves. The document travels through three gates, and any failure names *every* problem with its path (`bodies[1].shape.radius`) so it can be fixed in one pass:

1. **Syntax** — a schema rejects unknown keys, missing positions, malformed shapes.
2. **Semantics** — unique ids, joints that reference real bodies, no zero-area dynamic bodies that the engine would silently give a default mass.
3. **World** — the validated document becomes a running simulation at a fixed 1/120 s step, decoupled from the frame rate so the physics is the same on a laptop and a projector.

Current progress

Working today	Deliberately deferred
Rigid-body simulation (planck.js) at a fixed 1/120 s step	Forces beyond joints, gravity and contacts
Scene document with schema + semantic validation	Compound bodies, springs, friction models
Circles, boxes, edges; static and dynamic bodies	More joint kinds (revolute, prismatic, weld)
Distance joints with local anchors	Saving and sharing scenes (persistence)
Live force vectors: gravity, joint reaction, normal	Natural-language (*"a 2 kg block on a 30° ramp"*) input
Kinetic + potential energy, plotted over time	Multi-user gallery of published visualizations
Pluggable renderer contract (frame → backend)	Audio, 3D, mobile layouts
JSON editor with hot-reload; errors listed, never swallowed	

The sim and the renderer are already separate: a frame is a fresh plain value, so a second backend (WebGL, SVG, export) is one new file and no changes to the physics. The whole test suite runs headless — 57 checks covering parsing, validation, joint behaviour and the frame contract.

Roadmap

Phase	Goal	Done when
Now	Simulation, validation, force and energy display, JSON editing	Hackathon demo
1 — Express	More primitives: springs (Hooke's law integrated by hand), friction forces, applied forces, revolute/prismatic joints, compound shapes	A pendulum with a spring and a friction ramp is one short document
2 — Natural language	A prompt becomes a scene document. The model writes JSON only; the validator is the safety net, so an incoherent description is rejected with paths instead of rendered as nonsense	"A 2 kg block slides down a 30° ramp" plays, with no code
3 — Share	Named scenes, a URL per visualization, a gallery to browse and fork someone else's system	A teacher sends a link, a student opens it
4 — Classroom	Guided lesson mode: checkpoints, questions pinned to a moment in time, and a view students can drive themselves	A full lesson runs from one link

Objectives and why it matters

- **Minutes, not hours.** From a textbook sentence to a moving picture in under a minute — short enough to do it live, in front of a class.
- **Robust on stage.** A demo that says *why* it cannot run is worth more in a classroom than one that shows a plausible lie. Every invalid input is rejected with a readable path; the previous scene keeps playing.
- **Open and inspectable.** The scene is data, the physics is a documented engine, the source is GPL-3.0. A teacher can hand the document to a colleague, and a student can see there is no magic.
- **A foundation, not a toy.** Forces and energy are exposed through one renderer-agnostic contract, so sharing, AI authoring and new backends are additive work rather than rewrites.

Technical status at time of writing — 57 passing tests, headless, no network calls; TypeScript strict; scene schema and validation complete for the current primitives; rendering decoupled from the simulation. **newton.kaarlo.dev**